Md Abrar Jahin, CSCA™

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Citizenship: Bangladesh

GitHub
LinkedIn
Personal Website

Google Scholar
ResearchGate
ORCiD

Education

Khulna University of Engineering & Technology — Khulna, Bangladesh
B.Sc. Eng. in Industrial & Production Engineering — *Jan 2019 – Mar 2024*
CGPA: 3.83/4.00 (Top 5% of class)
• Dean's award: 2018-2019, 2019-2020, 2020-2021
• Thesis title: Supply Chain Backorder Prediction Using Interpretable Hybrid Quantum–Classical Neural Network [Thesis Presentation] [*Supervisor*: Dr. Md. Saiful Islam]
• Developed the first-ever LaTeX template for B.Sc. Undergrad Thesis of KUET [Template]

Research Interests

† Machine Learning & Deep Learning
• *Energy Efficient Deep Learning, Explainable AI (XAI), Kolmogorov-Arnold Networks (KAN), Physics-Informed Neural Networks (PINN), Reinforcement Learning (RL), Natural Language Processing (NLP)*
† Quantum Computing
• *Quantum Machine Learning*
† Uncertainty Quantification
• *Conformal Prediction*
† Comparative Genomics

Research Experiences

❖ **Visiting Researcher (VR)** — *Mar 2024 – Present*
Physics and Biology Unit,
Okinawa Institute of Science and Technology Graduate University (OIST), Japan
Supervisor: Prof. Jonathan Miller [BS (Yale); PhD Biology (Cambridge); PhD Physics (Caltech)]
Research project: Evolution of Strongly Conserved Sequence [Code Repository]
Google Knowledge Panel of Md Abrar Jahin

❖ **Visiting Research Student (VRS)** — *Feb 2023 – Feb 2024*
Physics and Biology Unit, OIST, Japan
Supervisor: Prof. Jonathan Miller
Research project: Evolution of Strongly Conserved Sequence [Certificate]
Collaborators: Dr. Lucia Žifčáková, Dr. Priscila Do Nascimento Biller, Dr. Zdenek Lajbner, and Dr. Reuven Pnini

- Critically analyzed and visually represented all potential combinations of inter-gap segments (IGS), ancestral repeats (ARs), and contiguous mismatched ARs in human/mouse and human/gorilla genome alignments, focusing on both DNAs and repetitive sequences.
- Successfully replicated the findings of the neutral indel model proposed by Lunter, Pointing, and Hein (2006).

❖ **Researcher** — *Mar 2023 – Present*
Advanced Machine Intelligence Research Lab (AMIRL),
American International University-Bangladesh (AIUB), Bangladesh

Research Affiliations:

- ♦ Department of Natural Language Processing and Computational Linguistics
  Supervisor: Dr. M. F. Mridha (Associate Professor, Dept. of CS, AIUB)
- ♦ Department of Deep Learning and Computer Vision
  Supervisor: Jungpil Shin (Professor, Division of Information Systems, School of Computer Science and Engineering, University of Aizu, Japan)
- ♦ Collaborators: Prof. R. Simon Sherratt (IEEE Fellow), Dr. Nilanjan Dey, Dr. Yuichi Okuyama, Dr. Zeyar Aung

❖ Research Lead *May 2022 – Mar 2023*

Research Camp 02, Scholarship School BD, Bangladesh
Supervisor: Dr. Mohammad Arafat Hussain (Post-doctoral Research Fellow at Image, Informatics & Intelligence Research Lab, Harvard Medical School; PhD in Biomedical Eng., UBC Canada; MASc in Biomedical Eng., UBC)
Research project: Ultrasound-Based AI for COVID-19 Detection

- ■ Led the research team of 17 fellow researchers on a project titled "Ultrasound-Based AI for COVID-19 Detection: A Comprehensive Review of Public and Private Lung Ultrasound Datasets and Studies".
- ■ Contributed to writing the original manuscript, software implementation, data curation, and successfully submitting to a journal.

❖ Research Intern (RI) *Oct 2021 – Mar 2022*

Physics and Biology Unit, OIST, Japan
Supervisor: Professor Jonathan Miller

- ■ Awarded with full-funded scholarship with a daily allowance of JPY 2400 per working day (taxable) [Offer Letter] [RI Agreement] [Internship Certificate]
- ■ Conducted research on perfectly conserved sequence (PCS) length distributions of UCSC 44 pairwise genome sequences.
- ■ Analyzed quantile kurtosis of PCS lengths proposed by Nash and Lenhard (2018) and identified a 'knee' in the PCS distributions of the heavy-tailed region.
- ■ Collaborated with Dr. Zifcakova to track erroneous out-of-bound PCS coordinates generated by Nash and Lenhard (2018), utilizing R and Bedtool, and resolved issues using Python 3.10.
- ■ Enhanced efficiency by optimizing Nash and Lenhard's 3 R scripts for PCS generation, quantile kurtosis analysis, and genomic regulatory blocks (GRBs), reducing time and memory complexity. Successfully reproduced PCS coordinates following UCSC format.
- ■ Demonstrated proficiency in shell scripting, programming, and software installation on the HPC Deigo cluster.

❖ Research Intern *May 2021*

UiT - The Arctic University of Norway

Supervisors: Professor Aleksander Pedersen, Professor Rune Dalmo, Ghada Bouzidi

- Conducted comprehensive Data and Statistical analysis on the Narvik road dataset using Python and MS Excel as part of the DIT4BEARs Smart Road project. [Internship Report] [Project Source Code] [Certificate]
- Designed, implemented, and evaluated machine learning models, including KNN, Logistic Regression, XGBClassifier, and Prophet. These models successfully identified road states (Dry, moist, wet, icy, snowy, and slushy), addressing the challenges posed by winter weather conditions in the Barren Euro-Arctic region.
- Pioneered the introduction of a novel safety metric and utilized Ridge, Lasso, Elastic Net, Linear Regression, and XGBRegressor to forecast its values.
- Effectively communicated project findings by presenting results at the final meeting, attended by 4 distinguished professors.

Honors and Scholarships

MIT Solve — 2024 Global Health Equity Challenge 2024
Founded SpecX - An XAI-Powered Web App for Sentiment-Driven Disease Profiling & Specialist Allocation [Solution]

Champion — CS50x Puzzle Day — Harvard University — Meta 2024
Led a 5-person international team by fostering diversity & inclusion (Bangladesh, USA, Morocco, and Pakistan) and solved advanced 9/9 puzzles (including Metapuzzle) [Certificate]

Dean's Award (3x) — KUET 2023
Received 3 Dean's awards in recognition of achieving annual GPAs greater than 3.75 out of 4.00 in three consecutive years of undergraduate classes

NASA Space Apps Challenge — Global Nominee 2023
Led a 5-person team and forecast geomagnetic storms using hybrid deep neural networks from satellite data — [Project] [30 seconds of glory video]

Finalist — HONDA Y-E-S Award 2023
Awarded for being among the top 15 Bangladeshi young engineering undergrad student scientists [Award & certificate]

Junior Research Fellowship — SPARRSO 2022
Nominated as the junior-most research fellow among the other excellent 55 undergraduate researchers for the project titled "Disaster Damage Mitigation by Multispectral Remote Sensing Satellite Image Data Analysis: A Deep Learning Approach" [Project nomination] [Presentation video]

Qiskit Gold Level Translator — English to Bengali 2021
Translated 22101 and proofread 25375 words of IBM quantum computing language Qiskit's first-ever textbook collaborating with West Bengal and Bangladeshi Qiskit translator team of 36 members [Certificate]

Top 6 among 385 teams — Entrepret Season-2: Crafting Visions 2021
Developed a feasible and sustainable business canvas model showing the proper break-even point with financial breakdown, surveyed 160+ students for our Git and Jenkins integrated freelancing startup [Case solution] [Finalist]

Top 500 — Google Android App Developer Challenge 2021
Engineered a countdown timer app on *Android Studio* with *Jetpack Compose Beta* using Kotlin language [Source Code] [Google swags]

Global Champion — Smart Roads Hackathon 2021
Executed a 2-person team and devised an ML model to forecast winter road friction and was offered a 1-month research internship at UiT - The Arctic University of Norway [Source Code]

Winner — ISCEA Ptak Prize Global SCM Case Competition 2020
Led a 4-person team and achieved 70% scholarship for completing the course for the professional certification titled *Certified Supply Chain Analyst (CSCA)* [Case solution] [Certificate]

Champion — CS50x Puzzle Day (Fall) — Harvard University 2020
Spearheaded a 4-person international team (Bangladesh, the UK, Pakistan, and Mexico) and solved advanced 8/8 puzzles [Certificate]

Gold Honor — Ranked top 3% — IAAC 2020
International Astronomy & Astrophysics Competition [Solution][Final round certificate]

Champion — CS50x Puzzle Day (Spring) — Harvard University 2020
Led a 3-person international team by fostering diversity & inclusion (Bangladesh, Brazil, and India) and solved advanced 8/8 puzzles [Certificate]

International Asteroid Search Collaboration — NASA 2020
Administered a 4-person team and discovered 2 main belt asteroids by analyzing Pan-STARRS images using *Astrometrica* software [Certificate]

Gold Honor — Ranked top 5% — IYMC 2019
International Youth Math Challenge [Solution][Final round certificate]

Government Board Merit-based Scholarship (x4) 2010 – 2018
PSC (2010), JSC (2013), SSC (2016), HSC (2018) Board Exams

Publications

I have 19 citations according to Google Scholar as of September 20, 2024 (h-index = 3, i10-index = 1)

* *Denotes co-first authorship.*

Journal Articles

[J-1] **Jahin, M.A.**, Shovon, M.S.H., Shin, J., Ridoy, I.A., Tomioka, Y., & Mridha, M.F. (2023). Big Data - Supply Chain Management Framework for Forecasting: Data Preprocessing and Machine Learning Techniques. ***Archives of Computational Methods in Engineering***.

[J-2] **Jahin, M.A.**, Shovon, M.S.H., Islam, M.S., Shin, J., Mridha, M.F., & Okuyama, Y. (2023). QAmplifyNet: Pushing the boundaries of supply chain backorder prediction using interpretable hybrid quantum-classical neural network. ***Scientific Reports***, *13*(1), Article 1.

[J-3] **Jahin, M.A.**, & Talapatra, S. (2023). A Natural Language Processing-Based Classification and Mode-Based Ranking of Musculoskeletal Disorder Risk Factors. ***Decision Analytics Journal***, *11, 100464.*

[J-4] Ahmad, K.*, Islam, M.S., **Jahin, M.A.***, & Mridha, M.F. (2024). Analysis of Internet of things implementation barriers in the cold supply chain: An integrated ISM-MICMAC and DEMATEL approach. ***PloS ONE***, *19*(7), e0304118.

[J-5] Saha, A.K.*, **Jahin, M.A.***, Rafiquzzaman, M., & Mridha, M.F. (2024). Ergonomic Design of Computer Laboratory Furniture: Mismatch Analysis Utilizing Anthropometric Data of University Students. ***Heliyon***, *10*(14).

Under-review Journal Articles

[U-1] Shahriar, H.*, Islam, M.S., **Jahin, M.A.***, Ridoy, I.A., Prottoy, R.R., Abid, A., & Mridha, M.F. (2023). Exploring Internet of Things Adoption Challenges in Manufacturing Firms: A Delphi Fuzzy Analytical Hierarchy Process Approach. Submitted to ***PloS ONE***.

[U-2] Morshed, A., Shihab, A.A., **Jahin, M.A.***, Nahian, M.J.A., Sarker, M.M.H., Wadud, M.S.I.*, Uddin, M.I., Siraji, M.I., Anjum, N., Shristy, S.R., Rahman, T., Khatun, M., Dewan, M.R., Hossain, M., Sultana, R., Chakma, R., Emon, S.B., Islam, T., & Hussain, M.A.* (2023). Ultrasound-Based AI for COVID-19 Detection: A Comprehensive Review of Public and Private Lung Ultrasound Datasets and Studies. Submitted to ***Multimedia Tools and Applications***.

[U-3] **Jahin, M.A.**, Naife, S.A., Saha, A.K., & Mridha, M. F. (2024). AI in Supply Chain Risk Assessment: A Systematic Literature Review and Bibliometric Analysis. Submitted to ***International Journal of Machine Learning and Cybernetics***.

[U-4] **Jahin, M.A.**, Naife, S.A., Lima, F.T. J., Mridha, M.F., & Shin, J. (2024). Analyzing Male Domestic Violence through Exploratory Data Analysis and Explainable Machine Learning Insights. Submitted to ***Scientific Reports***.

[U-5] Rahman, M.M.*, **Jahin, M.A.***, Islam, M.S., Mridha, M.F., & Shin, J. (2024). Optimizing Container Loading and Unloading through Dual-Cycling and Dockyard Rehandle Reduction Using a Hybrid Genetic Algorithm. Submitted to ***European Journal of Operational Research***.

[U-6] **Jahin, M.A.**, Shovon, M.S.H., Islam, M.S., & Mridha, M.F. (2024). A Hybrid Transformer and Attention Based Recurrent Neural Network for Robust and Interpretable Sentiment Analysis of Tweets. Submitted to ***Scientific Reports***.

[U-7] **Jahin, M.A.***, Shahriar, A.*, Amin, M.A., & Mridha, M.F. (2024). MCDFN: Supply Chain Demand Forecasting via an Explainable Multi-Channel Data Fusion Network Model. Submitted to ***Heliyon***.

[U-8] **Jahin, M.A.**, Mridha, M.F., Aung, Z., Dey, N., & Sherratt, R.S. (2024). TriQXNet: Forecasting Dst Index from Solar Wind Data Using an Interpretable Parallel Classical–Quantum Framework with Uncertainty Quantification. Submitted to ***npj Artificial Intelligence***.

Research Internship Report

[R-1] **Jahin, M. A.**, & Krutsylo, A. (2021). DIT4BEARs Smart Roads Internship (arXiv:2107.06755). arXiv.

Conferences

[C-1] Žifčáková, L., & **Jahin, M. A.** (2023). Perfectly Conserved Sequences (PCS) Between Human and Mouse Are Significantly Enriched for Small-protein Coding Sequence. *Society for Molecular Biology and Evolution (SMBE)*, 2023. [Poster]

[C-2] Žifčáková, L., **Jahin, M. A.**, & Miller, J. (2022). Perfectly conserved sequences (PCS) between human and mouse are significantly enriched for small proteins. *Bioinformatics and Computational Biology Conference (BBCC), 2022*, 11. [Poster]

Manuscript in Preparation

[P-1] Žifčáková, L., & **Jahin, M. A.** (2024). *Perfectly conserved sequences (PCS) between human and mouse are significantly enriched for small proteins* [Unpublished manuscript]. Physics and Biology Unit, OIST

Teaching Experience

Intro to Programming with Python Jan 2022

Mini-Course Teacher, OIST, Japan

◇ Topics covered: Intro, Anaconda, variables, lists, strings, control structures [Course materials & details]

◇ Fellow Teachers: Dr. Nicholas Wardhana and Dr. Jeremie Gillet

Tutorials

Operations Research Jan 2023

- Developed and presented first-ever comprehensive Bengali online tutorials on Operations Research topics, facilitating 3rd-year IPE students.
- Topics covered: Simplex Method, Two-Phase, Big M, Graphical Sensitivity Analysis, TORA. [YouTube Playlist]
- Reference book: "Operations Research – An Introduction" by Professor Hamdy A. Taha.

Skills

Programming

Advanced and Proficient in: Python, C/C++, R, SQL, SAS, Data Structure and Algorithm, Object Oriented Programming

Familiar with: Kotlin

Machine Learning: Classical Deep Learning, Quantum Machine Learning (Qiskit, Pennylane), XAI, NLP, DASK: Parallel Computing, Tensorflow, Keras, PyTorch, IBM Watson

Data Analysis and Optimization: Microsoft Excel, IBM SPSS, Minitab, TORA, Beautiful Soup, Biopython, Bioconductor, NetworkX, OpenCV
High-Performance Scientific Computing: Deigo & Saion Cluster (OIST)
Operating System: Linux, Unix, Windows
Version Control: Git Bash, Github, GitLab
Product Design: AutoCAD 2021, SolidWorks 2022, Unity 2D
Referencing Software: Zotero, Mendeley
Reviewing & Bibliometric Analysis: Publish or Perish 8.0, Gephi, VOSviewer
Writing tools: LaTeX, Microsoft Word
Languages: Bengali (Native), English (Professional and fluent)

Leadership and Advisory Roles

KUET Research Society *Nov 2023 – May 2024*
Co-founder & President
• Served as an Executive Committee Member and President of the Industrial Engineering and Management Research Unit
• Supervising (as an alumnus) 6 groups of research students concentrating on ML-DL and computational fuzzy logic, fostering cross-departmental research collaboration
• Teaching (as an alumnus) scientific research methodology, research ethics, and journal article formatting meeting publication criteria and acquiring funding for publications

SpecX *Aug 2022 – Present*
Founder & AI Researcher
• Developed a hybrid DL model architecture, integrated XAI, topic modeling, telemedicine appointment, and personalized treatment
• Acquired funding from the University of Aizu (Japan) and AIUB (Bangladesh) to collect patients' data and conduct NLP research

Professional Affiliations

Peer Reviewer *Jul 2023 – Present*
Reviewed 8 journal articles verified by Web of Science as of September 20, 2024[WoS ResearcherID]
□ IEEE Access
□ Computers & Industrial Engineering
□ Journal of Contemporary African Studies
□ Journal of Multidisciplinary Healthcare
□ 7th European Conference on Industrial Engineering and Operations Management (Augsburg, Germany, July 2024)

Certifications

MITx: CTL.SC4x: Supply Chain Technology and Systems
MITx: 6.431x: Probability - The Science of Uncertainty and Data
MITx: CTL.SC0x: Supply Chain Analytics
MITx: 2.961.2x: Management in Engineering: Strategy and Leadership
HarvardX: PH125.1x: Data Science: R Basics
Delftx: UnixTx: Unix Tools: Data, Software and Production Engineering
TAUx, IsraelX: Unlocking Information Security: Part 1
ISCEA: Certified Supply Chain Analyst
Google: IT Technical Support Fundamentals
Google: Crash Course on Python
Google: The Bits and Bytes of Computer Networking
UCSanDiegoX: DSE200x: Python for Data Science
Georgia Tech: Speak English Professionally: In Person, Online & On the Phone
IBM: AI Chatbots without Programming

IBM: PY0101EN: Python 101 for Data Science
Microsoft: Introduction to Artificial Intelligence (AI)

Other Interests Lead guitar [YouTube], painting, ML hackathons [Devpost, Kaggle], business case solving [Slideshare].

Referees

Professor Dr. Jonathan Miller [Supervisor, OIST]
BS (Yale); PhD Biology (Cambridge); PhD Physics (Caltech)
Physics and Biology Unit
Okinawa Institute of Science and Technology Graduate University (OIST)
jm@oist.jp

Dr. Md. Saiful Islam [Undergrad thesis supervisor]
Associate Professor, Department of Industrial Engineering and Management
Khulna University of Engineering & Technology, Bangladesh
saifuliem@iem.kuet.ac.bd

Dr. M. Firoz Mridha [Supervisor, AMIRL]
Associate Professor (Senior Member, IEEE), Department of Computer Science
American International University-Bangladesh (AIUB)
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Dr. Nilanjan Dey [Supervisor, Human-in-the-Loop RL project]
Associate Professor (Senior Member, IEEE), Department of CSE
Techno International New Town, Kolkata, India
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Dr. Mohammad Arafat Hussain [Supervisor, Ultrasound-Based AI for COVID-19 Detection]
Research Fellow, Boston Children's Hospital, Harvard Medical School
PhD in Biomedical Eng., UBC, Canada
mohammad.hussain@childrens.harvard.edu

Md. Al Amin [Supervisor, MCDFN Project]
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Khulna University of Engineering & Technology, Bangladesh
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Professor Dr. Subrata Talapatra [Department head]
Department of Industrial Engineering and Management
Khulna University of Engineering & Technology, Bangladesh
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